

CropSage Final Database Storage Design

Production-ready MVP persistence contract for Texas farm profiles, 15 prewarmed agricultural sites, arbitrary farmer locations, provider evidence, deterministic scoring, scenarios, validation, agent and LLM conversations, and auditable application operations.

FINAL DECISION

Use 11 application tables in Supabase Postgres, PostGIS for farm geometry, one private Storage bucket for sanitized provider artifacts, and Git for the 22-crop catalog, Texas site manifest, scoring policy and source code.

Design status

Item	Decision
Scope	All 15 representative Texas agricultural sites plus any farmer-supplied Texas point or boundary.
Scoring authority	The deterministic engine alone calculates suitability, confidence, caps, factors and rank.
LLM authority	The LLM gathers missing inputs and explains validated results; it never calculates or changes scores.
Evidence providers	FortyGuard, NASA POWER, Open-Meteo and USDA SSURGO, with farmer and laboratory overrides where supplied.
Catalog	22 crops, catalog version 1.1.0 at design freeze; the exact version and SHA-256 hash are stored per run.
Security	All privileged database, provider and LLM operations are server-side. API keys and service-role credentials never reach the browser.

Final guarantees

- Every rendered recommendation can be reconstructed from an immutable farm profile, exact evidence bundle, catalog hash, scoring policy hash and engine version.
- Missing, stale or failed evidence remains explicit and can only lower confidence or block scoring.
- Provider requests retain endpoint, sanitized parameters, time range, location, spatial resolution, cache state, activity ID and artifact checksum.
- Follow-up scenarios and chats remain linked to the original assessment without mutating trusted state.
- The database stores no hidden reasoning, chain-of-thought, API secrets or unsanitized provider payloads.

1. Scope, components and trust boundaries

Supabase components

Component	Use	Boundary
Postgres	11 application tables, constraints, indexes, immutable run history.	Trusted structured state and metadata.
PostGIS	Farm point and optional Polygon or MultiPolygon boundary in SRID 4326.	Coordinates remain authoritative over labels.
Storage	Sanitized provider JSON, GeoJSON heatmaps, scenario exports and optional validated presentation artifacts.	Private bucket; signed URLs are generated temporarily and never saved.
Auth	Supabase Auth may own signed-in users; anonymous hackathon sessions remain server-managed.	auth.users is built-in and is not counted among the 11 tables.
Realtime	Optional subscription to run_events for visible progress.	No hidden reasoning or secret arguments.
Server API	Profile validation, provider adapters, scoring, validation, LLM calls and persistence.	Browser never receives provider, LLM or service-role keys.

Trusted versus presentation state

Class	Examples	Can affect scoring?
Trusted input	Versioned farm profile, validated EvidenceBundle, catalog and scoring configuration.	Yes, only through deterministic code.
Trusted output	Crop scores, caps, factors, confidence, validation report.	Output only; immutable after completion.
Operational state	Provider fetch status, cache metadata, retry count, run events.	No, except freshness and availability passed through validated evidence.
Presentation state	LLM explanation, chat message, validated display snapshot.	No. Never reused as scoring evidence.

What remains in Git

- data/crop-catalog/catalog.json and its JSON Schema.
- data/regions/texas_region_sites.json and its JSON Schema.
- data/scoring/crop_scoring_config.json, recommendation schema and deterministic source code.
- Migrations and contract tests. Each run saves the exact version, Git commit and hashes required for reproduction.

NO EXTRA TABLES

Do not create separate catalog, region-site, cache or scenario tables for the MVP. Git owns versioned reference data; provider_fetches owns cache metadata; recommendation_runs owns scenarios.

2. End-to-end persistence lifecycle

The saved chain

Stage	Writes	Required outcome
1. Session	assessment_sessions	Create a bounded anonymous or authenticated assessment context.
2. Profile	farm_profiles	Save a new immutable version for location, planting, irrigation and optional overrides.
3. Collection	provider_fetches + Storage	Reuse fresh cache or execute provider requests; retain safe lifecycle metadata.
4. Normalization	evidence_records	Create canonical facts with units, time, depth, space, provenance and quality.
5. Assembly	evidence_bundles + evidence_bundle_records	Save the exact bundle snapshot, completeness, freshness and record membership.
6. Scoring	recommendation_runs + crop_score_results	Persist one immutable run envelope and 22 auditable crop outcomes.
7. Validation	validation_reports	Reconcile totals, grounding and freshness; decide render_allowed.
8. Presentation	chat_messages + optional presentation snapshot	Explain only validated values and support follow-up questions.
9. Progress	run_events	Expose user-safe status for collection, scoring, validation and explanation.

Prewarmed and live paths

Path	Behavior
Representative site	Resolve site_id from the 15-site manifest, verify manifest hash, reuse fresh provider artifacts, assemble or reuse its bundle, then score the farmer profile.
Arbitrary Texas location	Normalize point or boundary, resolve region and timezone, derive location_hash, query cache, fetch missing or stale providers in parallel, assemble bundle, score and cache.
Provider failure	Save failed provider_fetches and events. Build a partial bundle only when allowed; otherwise return insufficient evidence without inventing values.
Follow-up scenario	Reuse unchanged profile and evidence, create a child recommendation_run with scenario_changes_jsonb, recalculate deterministic output and validate again.

Identity and immutability rules

- All primary keys are UUIDs generated server-side or by gen_random_uuid().
- External JSON identifiers such as profile_id, bundle_id and crop_id remain separate searchable columns.
- Completed profiles, bundles, recommendation runs, crop results and validation reports are append-only.
- Corrections create a new version or new run; they never overwrite historical evidence.
- Timestamps are timestamptz in UTC. Local interpretation always uses a stored IANA timezone.

3. Final table inventory

#	Table	Purpose	Primary relationships
1	assessment_sessions	Groups profile versions, baseline/scenario runs and chat follow-ups.	Has many profiles, runs, events and messages.
2	farm_profiles	Immutable farmer and resolved-location state.	Belongs to session; supersedes profile.
3	provider_fetches	API lifecycle, cache and sanitized artifact metadata.	May belong to profile; produces evidence records.
4	evidence_records	Normalized provider or farmer facts.	Belongs to provider fetch; reused through bundle join.
5	evidence_bundles	Exact normalized bundle snapshot supplied to scoring.	Belongs to profile; has many records through join.
6	evidence_bundle_records	Many-to-many membership and inclusion role.	Bundle to evidence record.
7	recommendation_runs	Baseline or scenario engine execution envelope.	Belongs to session/profile/bundle; may reference parent run.
8	crop_score_results	One auditable result for each of 22 crops.	Belongs to recommendation run.
9	validation_reports	Final deterministic authorization to render.	Exactly one final report per run.
10	run_events	User-safe operational progress and tool events.	Belongs to session and optionally run/fetch.
11	chat_messages	User and LLM conversation history and generation metadata.	Belongs to session; may reference profile/run.

Relationship map

Parent	Relationship	Child
assessment_sessions	1 to many	farm_profiles, recommendation_runs, run_events, chat_messages
farm_profiles	self-versioning	supersedes_profile_id
provider_fetches	1 to many	evidence_records
evidence_bundles	many to many	evidence_records through evidence_bundle_records
recommendation_runs	1 to 22	crop_score_results
recommendation_runs	1 to 1 final	validation_reports
recommendation_runs	self-referencing	parent_run_id for scenario comparison
chat_messages	optional reference	farm_profile_id and recommendation_run_id

4. Sessions and immutable farm profiles

Table 1 - assessment_sessions

Column	Type / constraint	Purpose
id	uuid PK	Stable assessment and conversation identifier.
owner_user_id	uuid null FK auth.users	Optional signed-in owner; built-in Auth table is not counted.
anonymous_token_hash	text null	Hash of server-issued anonymous session token; never store the raw token.
status	enum	active, completed, expired, archived.
active_profile_id	uuid null FK	Current immutable profile version.
active_run_id	uuid null FK	Last validated recommendation run.
structured_state_jsonb	jsonb	Pending slots, selected crop, last safe references and UI state; not scoring truth.
created_at / updated_at	timestamptz	Lifecycle timestamps.
last_activity_at / expires_at	timestamptz	Retention and session expiry.

Table 2 - farm_profiles

Field group	Required columns and rules
Identity	id; session_id; external_profile_id; profile_version; input_hash; supersedes_profile_id; created_at. Unique(session_id, profile_id, profile_version).
Geometry	farm_point geography(Point,4326) NOT NULL; farm_boundary geography(Geometry,4326) NULL; location_hash; farm_name and location_label nullable.
Resolution	location_mode; representative_site_id nullable; resolved_region_id; region_assignment_method; region_resolver_version; timezone; timezone_resolution_method; location_source.
Planting	planned_date or planned_month, never both; flexibility_days; requested_crop_id nullable.
Irrigation	irrigation_jsonb nullable: availability, reliability, method, source, well/canal capacity and notes. Unknown values remain unknown.
Overrides	soil_overrides_jsonb; current_soil_moisture_jsonb; recent_rainfall_jsonb; farmer_goal_jsonb.
Audit	source_metadata_jsonb; missing_fields_jsonb; profile_schema_version; completeness_notes.
REQUIRED INPUT	Only a valid location and planting date or month are required by the FarmProfile contract. Irrigation and optional farmer evidence may be absent; absence lowers confidence and is never converted into a favorable match.

Location resolution rules

- Representative site: persist site_id, parent_region_id, timezone, manifest version and manifest hash.
- Arbitrary location: persist resolver method/version and the resolved region and timezone.
- The original address label is descriptive only; coordinates are authoritative.
- Coordinate consistency between FarmProfile, provider requests and EvidenceBundle is validated within a documented tolerance.

TABLE 3

5. Provider collection and cache lifecycle

Table 3 - provider_fetches

Column group	Columns	Rule
Identity	id, request_group_id, farm_profile_id nullable, representative_site_id nullable, location_hash	request_group_id links all provider calls for one collection attempt.
Request	provider, endpoint, http_method, sanitized_parameters_jsonb, request_hash, cache_key	No secret headers, API keys or unsafe payload fragments.
FortyGuard	activity_id, poll_status, poll_count, filter_type, analytic_type, threshold_c, direction, granularity_m	filter_type=4 is the standard historical heatmap collection mode; exact parameters remain in JSONB.
Lifecycle	status, attempt_number, http_status, submitted_at, completed_at, fetched_at	status: queued, submitted, processing, completed, failed, timed_out, cancelled.
Cache	cache_state, cache_hit, expires_at, source_data_vintage	cache_state: live, fresh_cache, stale_cache, fallback, none.
Failure	error_code, safe_error_message, retryable	Errors are sanitized and never contain tokens or full sensitive responses.
Artifact	bucket_name, object_path, sha256, content_type, size_bytes, provider_timestamp, sanitization_status	Stable private Storage reference; never save a temporary signed URL.

Cache key contract

- Hash normalized provider name, endpoint, rounded point or canonical polygon, variable set, temporal window, time standard, spatial resolution and provider-specific parameters.
- FortyGuard cache keys include filter type, analytic type, threshold, direction, granularity and AOI.
- NASA keys include endpoint, community, variables, period and coordinate.
- Open-Meteo keys include model, timezone, current/recent/forecast variable sets and date range.
- SSURGO keys include coordinate or AOI, requested attributes, component strategy and depth intervals.

Initial configurable freshness defaults

Evidence	Fresh for	Reason
NASA climatology	180 days	Underlying climatological baseline changes slowly.
NASA daily history	30 days after final date	Completed historical dates are stable but artifacts may be regenerated.
SSURGO mapped soil	180 days	Soil survey attributes are slowly changing.
Open-Meteo current	30 minutes	Operational current conditions change quickly.
Open-Meteo recent history	6 hours	Recent rainfall and soil state need timely refresh.
Open-Meteo forecast	1 hour	Forecast runs are revised frequently.
FortyGuard historical heat	30 days	Completed historical windows can be reused.
FortyGuard current/forecast	1 hour	Operational heat signals require recent generation.

6. Normalized evidence and immutable bundles

Table 4 - evidence_records

Group	Columns	Notes
Identity	id, provider_fetch_id, canonical_variable, dedupe_hash	One normalized fact or compact series per record.
Value	value_numeric, value_text, value_jsonb	CHECK that an appropriate representation is populated; null means unknown, never zero.
Units	normalized_unit, source_field, source_unit, conversion_method	Canonical units follow EvidenceBundle 1.1.0.
Time	temporal_role, observed_from/to, valid_from/to, forecast_issued_at, source_timezone, normalized_timezone, aggregation_method	Separates climatology, history, recent, current and forecast evidence.
Space	support_geometry, spatial_support, spatial_resolution_m	Point, farm area, tile or map unit.
Depth	depth_top_cm, depth_bottom_cm, depth_basis	Required for SSURGO horizons and soil-moisture layers.
Quality	quality_status, freshness_status, quality_flags_jsonb, warnings_jsonb	Statuses remain visible to the scorer and validator.

Tables 5-6 - evidence_bundles and evidence_bundle_records

Table	Required fields
evidence_bundles	id; external_bundle_id; farm_profile_id; schema_version; bundle_hash; location_hash; representative_site_id; region_manifest_version/hash; assembled_at; status; provider_coverage_jsonb; completeness_percent; freshness_summary_jsonb; invalid_numeric_count; bundle_jsonb; validation_jsonb.
evidence_bundle_records	evidence_bundle_id; evidence_record_id; inclusion_role; inclusion_order; required_for_scoring. Composite primary key on bundle and record.

Bundle statuses

Status	Meaning	Scoreable?
assembling	Provider collection or normalization is incomplete.	No
validated	Contract, units, location and freshness checks passed.	Yes
partial	Some optional evidence is unavailable; completeness and missing fields are explicit.	Only if essential-evidence policy passes
stale	One or more required artifacts exceed freshness policy.	Normally no; disclosed fallback only
invalid	Schema, coordinate, unit or numeric validation failed.	No
failed	Assembly could not produce a usable bundle.	No
REPRODUCTION RULE The exact EvidenceBundle JSONB and SHA-256 hash are stored even though normalized facts are queryable separately. The scorer receives the immutable bundle snapshot, not a fresh reconstruction from mutable rows.		

7. Provider-specific evidence contract

Provider	Persisted request metadata	Normalized evidence retained
FortyGuard	AOI, filter_type=4, window, analytic type, threshold, direction, granularity, activity_id, polling lifecycle and response tiles.	TCM mean/min/max, tile count, exceedance hours, persistence hours, peak hour raw value and timezone status, threshold-specific crop linkage, map/GeoJSON artifact.
NASA POWER	Point, temporal endpoint, AG community, parameter list, date or climatology period and time standard.	Daily/monthly/annual T2M, T2M_MAX, T2M_MIN, precipitation, RH2M, wind, solar, PAR, root-zone wetness, ET energy, derived VPD, GDD and heat-stress days with interpretation warnings.
Open-Meteo	Point, timezone, forecast model, current/recent/forecast variable sets, past and forecast day counts.	Current conditions, recent rainfall, daily min/max temperature, precipitation, ET0, humidity, wind, solar radiation, soil moisture layers, soil temperature layers, forecast heat/frost and water-balance summaries.
USDA SSURGO	Point/AOI, map-unit and component selection method, requested horizons and depth intervals.	mukey/musym/muname, component, farmland class, texture and fractions, pH, drainage, hydric rating, root limit, available water storage by depth and horizon details.
Farmer/lab	Profile field, source, observation or test time, method and optional reference.	Irrigation, soil texture, laboratory pH, soil moisture, rainfall and farmer goals; explicit precedence over mapped values where policy permits.

Provider precedence and interpretation

- FortyGuard is the central local heat provider. NASA and Open-Meteo do not replace FortyGuard heat evidence.
- NASA supplies long-term and historical climate context, not field-scale measurements.
- Open-Meteo supplies operational current, recent and short-term forecast weather, including daily minimum and maximum values.
- SSURGO supplies mapped soil context. A valid laboratory pH or documented farmer observation may override the mapped value while retaining both records and their sources.
- Every derived metric identifies its input evidence_record IDs and derivation version.

FortyGuard threshold collection

For each site and active heat window, collect one TCM request plus exceedance and persistence requests for each unique scoreable crop threshold. Store a threshold-to-crop mapping so one provider result can support multiple crops without duplicate API calls. Informational-only or non-numeric thresholds are retained as warnings and do not receive numeric penalties.

8. Deterministic scoring and scenario persistence

Table 7 - recommendation_runs

Group	Columns
Identity	id, session_id, farm_profile_id, evidence_bundle_id, requested_by_message_id nullable, created_at.
Mode	evaluation_mode, requested_crop_id nullable, status, baseline_or_scenario, parent_run_id nullable.
Scenario	scenario_batch_id, scenario_key, scenario_label, scenario_type, scenario_changes_jsonb, assumptions_jsonb.
Versions	profile_version/hash; evidence schema version/hash; catalog version/hash; region manifest version/hash; engine version/Git commit; scoring policy version/hash; recommendation schema version.
Snapshots	engine_input_jsonb and engine_output_jsonb, both immutable.
Timing	started_at, completed_at, duration_ms, safe_error_code, safe_error_message.
Presentation	validated_presentation_jsonb nullable; populated only after render_allowed=true and never used as future input.

Table 8 - crop_score_results

Group	Columns
Identity	id, recommendation_run_id, crop_id, crop_name. Unique(run_id, crop_id).
Outcome	status, eligibility, hard_filter_reasons_jsonb, rank, suitability_score, recommendation_band.
Confidence	confidence_score, confidence_band, evidence_coverage_percent, confidence_inputs_jsonb.
Calculation	category_components_jsonb, factors_jsonb, comparison_metrics_jsonb, applied_caps_jsonb.
Explanation inputs	strengths_jsonb, risks_jsonb, warnings_jsonb, reason_codes_jsonb.
Provenance	evidence_record_ids_jsonb, missing_evidence_jsonb, supplementary_evidence_jsonb.
Scenario	baseline_score, score_delta, baseline_rank, rank_delta.
SUCCESSFUL RUN INVARIANT	A successful rank-all run has exactly 22 crop_score_results. Every factor keeps weight, availability, factor score, evidence confidence, weighted points, scoring use, reason and evidence references.

Scenario matrix support

Scenario type	Saved change
in_season_baseline	No change; parent_run_id is null.
out_of_season	Changed planned month/date.
no_irrigation	Irrigation availability set to no.
reliable_irrigation	Availability yes and reliability reliable.
missing_or_lab_override	Explicitly hidden optional evidence or supplied laboratory soil override.

The 15 locations times five scenarios form a scenario_batch of 75 deterministic runs. Static JSON and CSV exports are stored in Storage with a batch identifier, checksum and generation time.

9. Validation and user-safe operational events

Table 9 - validation_reports

Column	Type / content
id / recommendation_run_id	uuid; unique final report per run.
validator_version	Version or Git commit of deterministic validator.
status	passed, passed_with_warnings, failed.
schema_checks_jsonb	Input/output schema results.
reconciliation_jsonb	Weights, totals, ranks, caps and 22-crop checks.
grounding_jsonb	Catalog crop IDs, thresholds, regions and evidence links.
freshness_jsonb	Provider coverage and expiry assessment.
errors_jsonb / warnings_jsonb	Structured machine-safe diagnostics.
render_allowed	boolean NOT NULL; only true after all blocking checks pass.
validated_at	timestampz.

Table 10 - run_events

Column	Purpose
id, session_id, run_id, provider_fetch_id	Relate the event to the user-visible workflow.
sequence_number	Stable ordering within a session or run.
event_type / tool_name	profile, provider, bundle, scoring, validation, llm or system event.
status	queued, running, completed, failed, skipped, cached.
started_at / completed_at / duration_ms	Operational timing.
cache_state	Live or cached classification shown safely to the user.
sanitized_arguments_jsonb	Approved non-secret argument summary.
user_safe_summary	Short progress message suitable for UI display.
safe_error_code / safe_error_message	Sanitized failure information.

Realtime policy

- The UI may subscribe to run_events scoped to its session.
- Events may state which tool ran, whether data was live or cached, and whether it succeeded.
- Events must never contain prompts, hidden reasoning, credentials, authorization headers or complete unsanitized provider payloads.

TABLE 11

10. Agent and LLM conversation persistence

Table 11 - chat_messages

Column group	Columns	Rule
Identity	id, session_id, sequence_number, created_at	Unique(session_id, sequence_number).
Message	role, message_type, content	role: user, assistant, system_safe_summary, tool_status.
Trusted references	farm_profile_id, recommendation_run_id, validation_report_id	References trusted structured state; prose itself is not trusted input.
Structured context	structured_context_jsonb	Selected crop, unresolved questions and safe IDs only; no chain-of-thought.
Generation	model_provider, model_name, generation_id, prompt_template_version, prompt_template_hash	Supports OpenRouter or another server-side model without coupling scores to the model.
Metrics	input_tokens, output_tokens, latency_ms, finish_reason	Optional operational cost and latency tracking.
Safety	safety_status, error_code, redaction_status	Store only user-safe content.

Agent context assembly

Context supplied to LLM	Source	Trusted?
Farmer request and recent messages	chat_messages	Conversation only
Current normalized farm state	active farm_profiles row	Yes
Latest approved recommendation	recommendation_runs + crop_score_results	Yes
Validation authorization	validation_reports	Yes
Evidence summary and source timestamps	evidence_bundles + records	Yes
Catalog descriptions and limitations	Git catalog version/hash used by run	Yes
Previous LLM prose	chat_messages	No; useful for conversational continuity only

LLM BOUNDARY

The LLM may ask follow-up questions, choose approved tools, summarize evidence, compare validated crops and explain results. It may not calculate suitability, alter confidence, remove risk flags, create crop thresholds or authorize rendering.

No vector database for MVP

Do not add pgvector or embeddings for the hackathon. The active profile, validated run, evidence summaries and recent chat messages provide sufficient bounded context. Retrieval from an embedding store would add complexity without improving scoring authority.

11. Supabase Storage design

Bucket

PRIVATE BUCKET	provider-artifacts - server writes only; signed read URLs are generated on demand and expire quickly.
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Object path contract

Artifact	Stable object path
Provider JSON	regions/{site_id}/{provider}/{request_hash}.json
Arbitrary location JSON	locations/{location_hash}/{provider}/{request_hash}.json
FortyGuard map	heatmaps/{location_hash}/{activity_id}/{analytic_type}.geojson
Scenario JSON	scenario-exports/{scenario_batch_id}/scenario_results.json
Scenario CSV	scenario-exports/{scenario_batch_id}/scenario_summary.csv
Validated presentation	presentations/{run_id}/{version}.json or .pdf when explicitly produced

Artifact metadata saved in provider_fetches or the owning run

- bucket_name and object_path
- SHA-256 checksum, content type and size in bytes
- provider generation timestamp and application upload timestamp
- sanitization status and sanitizer version
- artifact schema version where applicable

Never store

Excluded	Reason
API keys and authorization headers	Secrets belong only in server environment variables.
Unsanitized provider payloads	May contain unnecessary metadata and increase accidental disclosure risk.
Temporary signed URLs	They expire; store the stable bucket/object reference instead.
Hidden reasoning or chain-of-thought	It is unnecessary, untrusted and not part of the product contract.
Unnecessary farmer PII	Coordinates and farm decision inputs are sufficient for the MVP.

12. Security, RLS and privacy

Access model

Actor	Allowed access
Browser	Calls the CropSage server API. It does not receive service-role, provider or LLM credentials.
Anonymous farmer	Accesses only the server-managed session associated with a secure HTTP-only session token.
Authenticated farmer	May access rows whose owner_user_id equals auth.uid(), if authentication is enabled.
Server application	Uses a protected service role for validated writes, provider cache operations and scoring persistence.
Public demo	May read only explicitly published, sanitized demo outputs through a safe server endpoint or read-only view.

RLS decisions

- Enable RLS on all 11 application tables.
- Deny direct anonymous inserts, updates and deletes. Anonymous workflows write through the server.
- Authenticated reads require session ownership through owner_user_id or an ownership join.
- Shared representative-site cache rows are not user-owned; expose them only through approved server functions or views.
- Storage policies deny public listing and direct writes. The server generates short-lived signed URLs after authorization.

Immutability

- Database triggers reject UPDATE/DELETE on completed farm profiles, validated bundles, completed runs, crop results and validation reports except controlled administrative retention jobs.
- Corrections create superseding profile versions or new runs.

Privacy and redaction

- Do not require farmer name, email, phone number or property ownership records.
- Sanitize provider parameters and errors before database insertion.
- Chat content is presentation state and follows session retention; it is not silently promoted to evidence.
- Logs and run events use user-safe summaries rather than raw prompts or payloads.

13. Constraints, indexes and integrity checks

Required extensions

- pgcrypto for gen_random_uuid() and hashing support.
- postgis for farm points, boundaries and spatial indexes.

High-value constraints

Entity	Constraint
farm_profiles	Exactly one of planned_date or planned_month; valid Texas coordinate for production scope; version greater than zero.
provider_fetches	Unique request_hash/cache_key for reusable completed artifacts where policy permits; no raw secret fields.
evidence_records	Valid unit; depth_top <= depth_bottom; appropriate value representation; freshness and temporal roles enumerated.
evidence_bundles	bundle_hash unique; completeness 0-100; validated status requires bundle_jsonb and passed validation.
evidence_bundle_records	Composite PK(bundle_id, record_id).
recommendation_runs	Scenario requires parent_run_id; baseline forbids parent_run_id; completed run requires engine output.
crop_score_results	Scores 0-100; rank 1-22; unique(run_id,crop_id); unique non-null(run_id,rank).
validation_reports	Unique recommendation_run_id; render_allowed false whenever blocking errors exist.
run_events / chat_messages	Unique sequence number within owning session or run.

Indexes

Index	Purpose
GIST farm_profiles(farm_point)	Nearest-location and Texas geographic queries.
GIST farm_profiles(farm_boundary)	Boundary intersection where supplied.
provider_fetches(cache_key, status, expires_at)	Fast fresh-cache lookup.
provider_fetches(location_hash, provider)	Reuse provider evidence by location.
evidence_records(canonical_variable, valid_from, valid_to)	Retrieve time-aligned facts.
recommendation_runs(session_id, created_at desc)	Latest run and follow-up context.
crop_score_results(run_id, rank)	Ranked cards and comparisons.
run_events(session_id, sequence_number)	Realtime ordered progress.
chat_messages(session_id, sequence_number)	Ordered conversation context.

14. Retention, freshness and deletion

Data	MVP retention	Deletion behavior
Assessment sessions	30 days after last activity unless marked demo or retained by signed-in user.	Expire first; hard delete in scheduled maintenance.
Farm profiles and chat	Follow owning session retention.	Cascade from deleted session, except published sanitized demo profiles.
Recommendation runs/results/validation	90 days minimum for hackathon audit; longer for published demo runs.	Delete with owning session unless explicitly archived as sanitized benchmark.
Provider fetch metadata	90 days after artifact expiry; completed shared caches may be retained while referenced.	Preserve rows referenced by retained evidence bundles.
Evidence records and bundles	At least as long as any retained recommendation run references them.	Reference-aware deletion only.
Storage artifacts	Delete 30 days after expiry when no retained bundle or run references the checksum.	Scheduled reference check before object removal.
Run events	30 days after session completion.	Cascade with session/run.

Freshness is not retention

An expired cache artifact may remain stored for audit while being ineligible for a new recommendation. `expires_at` controls reuse; retention controls when the record and object may be deleted.

Deletion safety

- Never delete an artifact that is referenced by a retained `EvidenceBundle` or completed recommendation run.
- Shared cached provider evidence uses `SET NULL` or restricted foreign-key behavior rather than accidental session cascade.
- Material deletions are executed by an administrative retention job and logged outside user-facing run events.

15. Failure, retry and partial-evidence behavior

Situation	Persisted behavior	User result
FortyGuard processing timeout	Keep activity_id and processing/timed_out fetch row; next attempt polls the same activity before resubmitting unless provider guidance says otherwise.	Still processing or disclosed fallback; no duplicate blind submission.
Provider 4xx validation error	Save safe error code/message and failed status; do not cache as evidence.	Ask for corrected input or return insufficient evidence.
Provider 5xx/network failure	Retry within bounded policy; record each attempt and final state.	Fresh cache if available, otherwise partial/failed collection.
Stale cache	Retain artifact but mark stale; do not label as live.	Use only under disclosed fallback policy and lower confidence.
Coordinate mismatch	Bundle status invalid; store validation diagnostics.	Block scoring.
Missing optional evidence	Bundle status partial with explicit coverage and missing variables.	Renormalize available weights and lower confidence if scoring policy allows.
Missing essential provider/location	No scoreable bundle or run status insufficient_evidence.	No final score.
LLM failure	Validated deterministic output remains stored; save safe message error.	Render structured results without chat explanation.
CORE FAILURE RULE	Operational failure must never create a favorable score. Cached, partial, stale and missing states are distinct and are carried into confidence and validation.	

16. Project-phase storage coverage

Phase	Persisted artifacts	Database/storage support
1. 15-site manifest	Versioned Git manifest and verified coordinates.	Profiles/runs store site_id, manifest version/hash, region and timezone.
2. Generic evidence builder	Location-independent EvidenceBundle.	evidence_bundles, records and membership store exact normalized evidence.
3. Automated collection	Four provider JSON artifacts and fetch lifecycle.	provider_fetches plus private Storage paths/checksums.
4. Collect 15 sites	One request group and bundle per site/window.	collection_batch_id, representative_site_id and location_hash.
5. Generate scenarios	Five scenario definitions per site.	recommendation_runs parent link, scenario type/key/change snapshot.
6. Run 75 scenarios	Results JSON, summary CSV and validation.	75 runs, 1,650 crop results, reports and Storage exports.
7. Freeze scoring	Version 1.0.0 policy and engine commit.	Run-level versions and SHA-256 hashes.
8. Agent	Structured state, tool progress and chat.	sessions, run_events and chat_messages; LLM references validated runs.
9. Interface	Ranked cards, factor detail, heatmap and follow-ups.	Validated results, presentation snapshot and signed artifact access.
10. Deployment	Public demo, security and submission audit.	RLS, retention, immutable demo runs and server-only secrets.

Volume expectation

- Fifteen sites times five scenarios equals 75 recommendation runs.
- Seventy-five rank-all runs times 22 crops equals 1,650 crop_score_results.
- Provider request volume depends on unique FortyGuard scoreable thresholds; cached threshold results are shared across crops and scenarios.
- Large tiles and raw arrays remain in Storage; Postgres retains normalized scoreable facts and immutable snapshots.

17. Migration and implementation sequence

Order	Migration / implementation task	Acceptance check
1	Enable pgcrypto and postgis; create enums and helper functions.	Extensions enabled; timezone and enum values tested.
2	Create assessment_sessions and farm_profiles with RLS and spatial indexes.	Plainview and Poteet profiles insert as immutable versions.
3	Create provider_fetches and private provider-artifacts bucket policies.	One cached and one live request lifecycle can be represented safely.
4	Create evidence_records, evidence_bundles and evidence_bundle_records.	Existing Plainview EvidenceBundle round-trips with identical hash.
5	Create recommendation_runs and crop_score_results.	Current 22-crop output persists and reconstructs exactly.
6	Create validation_reports and immutability triggers.	Invalid run cannot set render_allowed=true.
7	Create run_events and Realtime policy.	Session receives ordered safe progress only.
8	Create chat_messages and LLM generation metadata.	Follow-up explanation references the validated run without changing numbers.
9	Implement cache lookup, collection batch and Storage checksum verification.	Poteet uses live/cache paths deterministically.
10	Seed the 15 sites and run 75 scenarios.	1,650 crop results, all validation reports and reproducible exports.
11	Freeze scoring version 1.0.0 and archive demo artifacts.	Catalog, manifest, policy, engine and validator hashes recorded.
12	Run security and end-to-end tests.	RLS isolation, no client secrets, expired sessions, provider failure and LLM failure verified.

Contract tests required before launch

- Coordinates and timezone align across profile, fetches, evidence and run.
- All 15 site IDs resolve to valid catalog regions and manifest hashes.
- A successful run contains 22 unique crop IDs and ranks 1-22.
- Suitability and confidence remain within 0-100 and separate.
- Missing evidence is never represented as zero or compatibility.
- Hard caps, planting windows, regional support and irrigation behavior are deterministic.
- Repeated identical inputs produce the same engine output hash.
- Only a passed validation report can authorize rendering or LLM explanation.
- Chat and run events contain no secrets or hidden reasoning.

18. Final decisions and exclusions

Decisions

Area	Final decision
Database	Supabase Postgres with 11 application tables.
Spatial	PostGIS geography Point plus optional Polygon/MultiPolygon, SRID 4326.
Files	One private provider-artifacts bucket with stable paths and SHA-256 checksums.
Reference data	Crop catalog, Texas site manifest and scoring configuration remain schema-validated in Git.
Scoring	Pure deterministic engine; immutable input/output snapshots and 22 queryable crop rows per rank-all run.
Validation	Deterministic final gate controls render_allowed.
Agent	One bounded agent using structured tools and stored safe state.
LLM	Server-side OpenRouter-compatible generation; explanation only, with messages and model metadata retained.
Chat	Persisted in chat_messages; never trusted as scoring input.
Security	Server-only privileged access, RLS on all tables, private Storage and no browser secrets.

Explicit exclusions

- No RNN or machine-learning forecast is required for the MVP database contract.
- No pgvector, embeddings or general-purpose knowledge-base table.
- No separate catalog, region-site, cache, scenario or model-output-as-truth table.
- No yield guarantee, irrigation prescription or hidden agronomic inference.
- No API keys, credentials, temporary signed URLs, unsanitized payloads or chain-of-thought.

READY FOR MIGRATION
DESIGN

This document is the final storage contract for the CropSage hackathon implementation. Step 3 provider adapters, the Supabase SQL migrations and the agent application should conform to these table, artifact, provenance, security and trust boundaries.

Internal contract sources

FarmProfile schema 1.0.0; EvidenceBundle schema 1.1.0; recommendation schema 1.0.0; crop catalog 1.1.0; scoring configuration 1.0.0-provisional; Texas 15-site manifest 1.0.0; current generic location evidence builder and deterministic scoring implementation.